

LAURENTIAN FAN BUOY 44141, Canada

WMO: 996260

Lat: 43.000N

Lon: 58.000W

Elev: 3

StdP: 101.29

Time Zone: -4.00 (W04)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.6	-2.2	N/A	N/A	N/A	N/A	N/A	N/A	19.7	6.0	18.5	4.6	9.4	300	0.649

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.3	24.6	N/A	23.9	N/A	23.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	6.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.5	15.5	13.9	DB	-4.5	24.9	2.2	1.5	-6.1	25.9	-7.4	26.8	-8.6	27.6	-10.3	28.7	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	6.2	5.0	5.1	6.6	9.9	14.4	19.2	20.7	18.8	15.2	11.8	8.5		
	DBStd	6.40	3.78	4.32	4.03	3.00	2.96	2.96	2.36	1.79	2.27	3.07	3.37	3.86		
	HDD10.0	683	126	149	162	109	39	2	0	0	0	2	19	76		
	HDD18.3	2544	376	374	412	351	263	124	17	2	22	103	196	306		
	CDD10.0	1349	8	9	8	8	34	133	286	332	263	164	74	29		
	CDD18.3	168	0	0	0	0	0	6	44	76	34	8	1	0		
	CDH23.3	121	0	0	0	0	0	1	37	69	14	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	6.6	8.1	8.0	7.5	6.3	5.3	5.0	4.9	5.0	6.0	7.4	7.4	8.2		
(15) Precipitation	PrecAvg	1583	154	124	115	90	94	114	125	139	115	151	172	189		
	PrecMax	2029	246	189	177	180	171	235	258	297	349	308	274	299		
	PrecMin	1214	60	70	58	22	18	38	41	40	22	27	83	90		
	PrecStd	209	42	33	31	36	39	53	53	59	70	67	61	57		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.7	17.5	17.8	17.5	19.5	22.8	25.1	25.6	24.7	22.6	21.1	18.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	15.5	15.6	15.7	15.2	17.7	21.4	24.4	24.7	23.7	21.7	19.7	17.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	14.0	14.1	13.8	13.3	16.0	20.1	23.6	24.2	22.8	21.0	18.4	16.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	12.4	12.3	11.7	11.2	14.1	19.2	22.9	23.6	22.2	19.9	17.0	14.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	5.6	4.9	3.7	3.4	3.0	2.4	2.3	2.7	3.5	4.3	5.5		
		MCDBR	9.0	8.4	8.8	7.7	5.8	3.1	2.8	2.9	3.4	4.4	6.2	8.1		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.338	0.332	0.351	0.379	0.388	0.406	0.432	0.398	0.378	0.363	0.339	0.327		
	taud		2.421	2.483	2.448	2.394	2.411	2.396	2.374	2.454	2.488	2.504	2.514	2.470		
	Ebn at Noon		789	868	892	888	886	867	840	859	853	821	788	769		
	Edh at Noon		78	86	100	114	115	117	118	106	95	83	71	68		
(44) All-Sky Solar Radiation	RadAvg		1.17	1.87	3.16	4.32	5.16	5.41	5.52	5.14	4.26	2.66	1.60	1.02		
	RadStd		0.10	0.19	0.28	0.35	0.37	0.47	0.49	0.35	0.34	0.21	0.19	0.09		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page